8 November 2026

Dear Editor,

Thank you, and thank the three Reviewers, for thoughtful review of our manuscript. We appreciate the feedback and have toned down the conclusions, and scope, of the paper to better indicate that we conducted a case study. The title is now "Do fault zones route road-salt-contaminated groundwater to streams? A Hudson Valley, NY Case Study".

There were small changes to the presentation of the results. To better address/resolve reviewer comments on site three we split the analysis based on the days the samples were taken. This provides more confindence in our responses.We could not collect new field data for this revision, so requests that need new measurements (discharge, tracers, repeated cross-channel sampling) are handled as limitations and future work.

Reviewer comments are in *gray italics*; our responses are in black. A reference such as {L135} is a line number in the highlighted-changes manuscript. latexdiff does not highlight tables read from separate files; Table 1, Table 2 and the Appendix A table are new or replaced, with captions marked.

Sincerely,

Tadd Bindas and Zion Klos

## Response to the Editor

### Comment E.1

*Overall, I consider the manuscript relevant and potentially suitable for publication, and therefore recommend publication provided that the points raised above are adequately addressed.*

**Response:** Thank you. We have responded to every point the Reviewers raised, as detailed below. The first author's affiliation is now Marist University, where the work was done.

## Response to Reviewer 1

### Comment R1 (general)

*The topic is novel and opens a useful and interesting discussion to associate structural geology directly to the freshwater salinization problem. The writing is generally clear and good. My main concern is that with the current structure, the data and statistics presented do not support the strength of the claims. The SC steps coincide with mapped faults at two crossings. However, the paper still does not show that the faults cause these steps, that the faults connect to a highway, or that road salt rather than another source is responsible. That's why I'd recommend to reframe the work as a hypothesis-generating case study, discuss alternative sources and the study's limitations explicitly in the text, clarify the maps and temper the conclusions.*

**Response:** Thank you for the feedback. To improve the paper's results/presentation we split site three based on the days we sampled. Only one of the two crossings the Reviewer mentions still shows a significant increase, and change-point analysis shows it to be a gradual rise rather than a step at the fault. Our data show where conductance changes along the channel, not that faults cause the changes, that they connect to highways, or that road salt rather than another source is responsible. We followed each recommendation: the title is now the question the study poses, alternative sources and limitations have their own sections (4.2 and 4.4), all maps are revised, and the conclusions are tempered. The Introduction now ends: "We present this work as a hypothesis-generating case study: a low-cost survey of this kind can show whether the predicted pattern is visible, and what a conclusive test would require, but we do not claim that it can, by itself, demonstrate that faults transport road salt." {L135} The Discussion opens: "Our survey offers little support for the hypothesis that fault zones route road-salt-contaminated groundwater to Wappinger Creek." {L434}

### Comment R1.1

*The Introduction reaches geology, the paper's central argument, only after long sections on impacts and the global salinization syndrome*

**Response:** The Introduction now moves from road salt and its groundwater legacy straight to faults: "Fault zones are one candidate control." {L91} We shortened the ecological-impact material and reduced the salinization syndrome to a single sentence that leads into the fault paragraph by naming geology and flowpaths among its state factors, beginning "Salinization is increasingly viewed as a coupled process" {L76}.

### Comment R1.2

*In the Introduction soil (not relevant here when looking specifically for chloride) and groundwater timescales are mixed together*

**Response:** Thanks for the comment. We removed the sentence on soil persistence (2.5–5 months) and kept only the groundwater storage and release argument: "Much of the applied chloride accumulates in groundwater and is released to streams over years to decades" {L71}.

**Comment R1.3**

*The three testable predictions have no supporting references*

**Response:** Thank you. Each prediction now has references (Harte & Trowbridge, 2010; Hodgins et al., 2024, 2026; Miller et al., 2014; Mullaney et al., 2009). For example, "dense longitudinal surveys have been used to locate point groundwater inputs of road salt to streams in this way" {L120}; the second prediction applies "when groundwater-fed baseflow supplies much of the streamflow at low flow" {L123}; and chloride-to-bromide ratios "are widely used to distinguish deicing salt from other chloride sources" {L124}.

**Comment R1.4**

*The fault hydrogeology rests on a karst analog and a study of unconsolidated aquifers, not on work in metamorphic bedrock (e.g., Caine et al., 1996; Bense et al., 2013).*

**Response:** Thank you for these references. The framing now rests on the fault-zone architecture model of Caine et al. (1996), the review of Bense et al. (2013), and studies of structurally controlled flow in fractured metamorphic bedrock (Manda et al., 2008; Kim et al., 2014), cited in Sections 1, 2.1 and 4.3 (for example {L97}). We removed the study of faults in unconsolidated sedimentary aquifers (Bense et al., 2008). We also removed the karst example (Ferrill et al., 2025).

**Comment R1.5(1)**

*A longitudinal survey of specific conductance as in the paper is a reasonable, low-cost way to screen for groundwater inputs. It can show where SC changes, but it is difficult to attribute those changes specifically to faults or to road salt so I'd recommend being careful in state that, more specifically: (1) There are no discharge data and no source tracers*

**Response:** Thanks for the comment. The survey is now described as a screening result {L496}, and the Limitations section (4.4) states the gaps: "We have no discharge measurements, so we cannot estimate the volume of any groundwater input, convert conductance to load, or separate dilution from addition." {L630} "We have no source tracers, so we cannot attribute salt to roads, wastewater or other sources." {L632} Paired discharge measurements, chloride-to-bromide ratios and groundwater age tracers are identified as the next step in Section 4.4 {L655}.

**Comment R1.5(2)**

*(2) There is only one chloride transect, measured with an ion-selective probe*

**Response:** Correct, there was only one pass done in 2020. The data are a single transect of 18 main-stem samples plus one side-tributary sample, but the Methods now name the sensor, a YSI Pro Series 1005 chloride ion-selective electrode {L223}, and give its accuracy: "The manufacturer gives an accuracy of ±15% of reading or ±5 mg/L, whichever is greater" {L224}. At our concentrations (about 14–50 $\mathrm{mg\,L^{-1}}$) that is ±5 to ±7.5 $\mathrm{mg\,L^{-1}}$, so only the larger differences along the transect

exceed it. We also state that "Because it is a single transect from a different year than the conductivity survey, it cannot confirm the conductivity pattern" {L228}, and list both the single transect and the measurement uncertainty among the limitations. See also R2.7.

**Comment R1.5(3)**

*(3) Comparing zone means with t-tests cannot show whether a change is abrupt. Change-point analysis or segmented regression of SC against distance would test this directly and could be recommended*

**Response:** This was a useful suggestion and we implemented it. For each single-day transect we fitted continuous piecewise-linear regressions of conductance on downstream distance with zero, one or two breakpoints chosen by BIC, and checked them with a penalized change-in-mean analysis (PELT) {L276}. The result: "Segmented regression and a change-in-mean check on each single-day transect (Figure 5) find no specific-conductance breakpoint at a fault contrast boundary or at a point where the digitized trace crosses the transect." {L381} At Site 2 a single breakpoint lies about 1.6 km upstream of the Contrast B boundary {L383}, after which conductance rises steadily (the fitted slope goes from 0.1 to 1.6 $\mu$S cm$^{-1}$ per km): "Contrast B is therefore better described as part of a gradual downstream ramp than as an abrupt step at the interpreted fault position." {L389} No breakpoint is supported at Site 1 or on the 19 October Site 3 survey, and two marginally preferred breakpoints on the upper 26 October transect lie away from any trace. On the lower 26 October transect (Contrast E) the only breakpoint sits at the three-point minimum segment the method allows, about 520 m upstream of the zone boundary and 123 m from the nearest mapped trace {L401}, whose position is itself uncertain (see R2.6). Changes in conductance are thus not concentrated at the mapped faults in this survey, and we revised the conclusions accordingly. Figure 5 is new.

**Comment R1.6**

*Table 1 lacks standard deviations, degrees of freedom, confidence intervals, and the Mann-Whitney results. I'd add that to help the reader interpret the results*

**Response:** Table 1 now reports date, zone sample sizes, zone means ± standard deviation, the difference in means with its 95% confidence interval, the Welch $t$ statistic, Welch–Satterthwaite degrees of freedom, $p$-value, and the Mann–Whitney $U$ statistic and $p$-value for each contrast and survey day. The Results introduce them: "Table 1 gives each contrast with zone means ± SD and the number of points $n$" {L292}.

**Comment R1.7**

*The reported t values imply a Zone 3-2 standard deviation larger than the contrasts being tested*

**Response:** You are correct, and your observation led us to split the measurements within our third zone definition. Zones 3-1, 3-2 and 3-3 each combined measurements from two survey days (19 and 26 October 2018). The Results now document the offset: "Eleven 19 October points have a 26 October point within 120 m, and the pairs differ by 23.5–29.7 $\mu$S/cm (mean 27.8, SD 2.2)" {L359}. Two measurements 18 m apart at the seam between the surveys read 332.0 and 355.5 $\mu$S cm$^{-1}$ {L360}. Within a single day the Site 3 zone standard deviations are 0.4–2.6 $\mu$S cm$^{-1}$; pooling inflated them to 12.6–15.8 $\mu$S cm$^{-1}$ {L357}, which is what the large $t$ values reflected. Our earlier statement that every contrast was computed within a single day's data was incorrect.

We now compare zones only within a single survey day: "Zones are compared only within a single survey day." {L255} Contrast E falls from +18.0 to +1.6 $\mu$S cm$^{-1}$ (95% CI −0.3 to +3.4; Welch $p = 0.090$), which is "not significant and within the instrument's stated accuracy" {L340}. We report the artifact in full: "When we pooled the two survey days, as a first analysis of the Site 3 data naturally would, Contrast E appeared as a rise of +18.0 $\mu$S/cm ($p < 0.001$), larger than any other step in the survey. However, this value is an artifact of the pooling" {L350} Sections 4.4 and 5 use the case as a pitfall to avoid {L655}.

**Comment R1.8**

*Spatial autocorrelation between samples 100 m apart is acknowledged but not handled, so the p-values are likely too small*

**Response:** Agreed. We now estimate the lag-1 autocorrelation along each transect, convert it to an effective sample size for each zone, and repeat the Welch tests with it {L269}. The adjustment raises $p$ wherever residuals are positively autocorrelated (Contrasts B, D, E, and C on 26 October). Contrast B stays significant: "Allowing for autocorrelation along the channel ($n_{\text{eff}} = 4.6$ and 3.7), the adjusted $p$ is 0.017" {L318}, against $p < 0.001$ unadjusted, while Contrasts D and E move further from significance ($p = 0.28$ and 0.24). With three to eleven points per zone the autocorrelation is poorly determined, so "we treat the adjusted $p$-values as a sensitivity check rather than a correction" {L274}. The change-point analysis (R1.5(3)) is a complementary test that does not treat samples as independent replicates.

**Comment R1.9**

*The z-score normalization that was presented on the Methodology has no effect on within-site tests*

**Response:** Thanks for the feedback. All tests were carried out on specific conductance in $\mu$S cm$^{-1}$; the normalization described in the Methods was left over from an earlier version of the analysis. We removed it and its equation, so the only remaining equation is the Welch $t$ statistic, a test of the difference in mean specific conductance "in $\mu$S/cm, upstream and downstream of each feature" {L242}. The site maps (Figures 2–4) now color points by specific conductance in $\mu$S cm$^{-1}$ as well.

**Comment R1.10**

*The explanation for the non-significant Contrast D is incorrect if you consider that two other contrasts have the same sample-size split*

**Response:** You are right: in the submitted analysis Contrasts C and E had the same 17/6 split, so unequal sample sizes cannot explain the result at Contrast D, and we removed that explanation. Within a single survey day Contrast D is a small, non-significant decrease: "It is a decrease of −2.8 $\mu$S/cm (336.3 ± 2.2, $n = 3$, to 333.4 ± 1.9, $n = 11$; 95% CI −7.4 to +1.7; $p = 0.140$)." {L337} We now say plainly: "Contrast D, at a fault with no tributary, shows no increase." {L468} The 2020 chloride peak lies near the fault points at Contrast D (see R2.7), but it rests on one sample from a different year, so we describe it as an observation worth testing, not as support.

**Comment R1.11**

*Some steps, such as 4.3 $\mu$S/cm at Contrast B, are close to the uncertainty of temperature-compensated SC*

**Response:** Agreed. The Methods now state the instrument accuracy: "the Orion meters compatible with this cell specify an accuracy of ±0.5% of reading ±1 digit, about ±1.7 $\mu$S/cm at 310 $\mu$S/cm. A calibration offset shared by all readings on one day cancels when zones are compared within that day, while random reading error does not." {L180} All Site 2 measurements were taken on one day with the same instrument and a narrow water-temperature range (15.4–15.8 °C), and the within-zone standard deviations (1.1 and 1.9 $\mu$S cm$^{-1}$) are smaller than the 4.3 $\mu$S cm$^{-1}$ difference. Contrast B is about 2.5 times the accuracy and Contrast E is within it {L494}. We nonetheless describe Contrast B as a small change near the limit of what the survey can resolve.

**Comment R1.12**

*The measured SC differences appear genuine and are within an expected range, but the interpretation goes beyond them. Other sources could fit the pattern at least as well as the faults so I'd be careful in how to state that specially considering that the highest-SC zones sit just beside US 44 / NY 82 (from the map), Site 2 runs beside developed land, A sewage treatment plant and septic systems lie upstream (according to the same study site evaluated in Kelly et al., 2008) and groundwater unrelated to faults can enter the creek anywhere.*

**Response:** Thank you for the feedback here. Section 4.2 now takes each alternative in turn, starting with the statement that several are "at least as plausible as fault-routed groundwater with the data in hand" {L504}.

*Road proximity.* We measured the distance from each sampling point to the nearest major road and report zone medians descriptively (245–927 m; Table 2), without attributing the contrasts to them. The Site 3 zones lie 245–425 m from the nearest major road, mainly US 44, with no monotone relation to the conductance pattern. At Site 2, "the downstream zone is farther from a major road than the upstream zone (median 726 versus 335 m), so proximity to a major road does not by itself explain Contrast B" {L519}, although it cannot be excluded.

*Developed land.* Site 2 runs beside US 44 and ends near the settled area around Titusville. We withdrew the argument that conserved forest lets us discount land-use inputs: "Sampling within conserved land does not let us discount such inputs, because the sources lie in the contributing area and not only along the survey line." {L532}

*Wastewater.* We located permitted facilities from EPA ECHO records. The Millbrook sewage treatment plant described by Kelly et al. (2008), whose study of the East Branch attributed about 4% of the sodium chloride input to sewage (septic systems and the plant together), is upstream of all three sites, about 2.4 km along the channel above Site 3. As a continuous upstream source it would raise baseline conductance but not by itself create a step at a zone boundary. More relevant to Contrast B, "two small private plants, Stratford Farms and Ennis Mobile Home Park, have listed locations 240–370 m from Zone 2-1" {L545}, upstream of the Site 2 breakpoint. Their outfalls and flows are unknown, so Section 4.2 names them as a plausible, untested alternative source. The Titusville plant lies just downstream of the last Site 2 survey point, so it cannot affect Contrast B, and the Valley Dale plant discharges to a tributary about 2 km away. The facilities are shown in Figures 1 and 3. Our data cannot distinguish these sources from fault-routed groundwater; that would need tracer and discharge measurements.

**Comment R1.13**

*Several statements overreach, such as "strong initial evidence" and "this newly observed process". Section 4.4 says all three crossings point to a link with fault zone transport, al-*

*though Contrast D shows a non-significant decrease. Recommending different deicers (in the Conclusions) wherever a road crosses a mapped fault is premature on three crossings in one lithology. The Discussion also offers little comparison with other longitudinal salinity surveys or with fault hydrogeology in metamorphic rocks.*

**Response:** Agreed on all points. We removed or rephrased the overreaching statements, including those quoted. You are right that the original Section 4.4 said all three crossings point to a link; we removed that sentence, and the contrasts are now described in Section 4.1, where Contrast D "shows no increase" (see R1.10). The management discussion (end of Section 4.3) no longer recommends changes in deicing practice: "We do not claim that the evidence here justifies changing deicing practice on the basis of fault locations." {L617} Mapped faults are at most a candidate screening layer to be tested. We added comparisons with other longitudinal salinity surveys (Harte & Trowbridge, 2010; Hodgins et al., 2024, 2026; Orr & Boyer, 2026) and with fault hydrogeology in metamorphic rock (Section 4.3), the former beginning "Other longitudinal surveys locate salt-laden groundwater inputs more directly than ours." {L606}

**Comment R1.14**

*some references lack DOIs or report numbers and grouped citations are out of order.*

**Response:** Corrected. DOIs are added where they exist. Four entries (Budnik et al., 2010; Kelly et al., 2019; Kelting & Laxson, 2010; Adirondack Road Salt Reduction Task Force, 2023) are reports without DOIs; they carry report numbers or URLs. Grouped citations are now sorted and compressed.

## Response to Reviewer 2

**Comment R2 (general)**

*This paper uses field measurements of conductance and chloride concentrations to demonstrate systematic changes in stream salinity across several mapped high angle faults in New York. The purpose of the study was to test whether the locations of high angle faults correspond to abrupt increases in specific conductance values, which the authors argue would provide evidence that groundwater flow along high angle faults can contribute salt-contaminated groundwater at point locations along streams.*

*Overall, this study's focus is on an interesting and understudied aspect of streamwater salinity – the authors present preliminary data somewhat consistent with their hypothesis. However, the paper would be strengthened by additional caution with some of their claims; improvements in the clarity of their figures; and additional discussion of alternative interpretations of their data.*

**Response:** Thank you for this fair assessment. We agree the data are preliminary and only partly consistent with the hypothesis. The claims are scaled back throughout (see R1 general), every figure is revised as requested, and alternative interpretations are in Sections 4.2 and 4.4.

**Comment R2.1**

*Introduction. Since many of your contrasts end up being 'confounded' by tributaries, adding a section to the introduction to discuss how tributaries typically influence downstream conductance would set you up to contrast the tributary vs fault boundaries you discuss in your*

*results. Perhaps the tributary dilution effect would be greater if not for the addition of salt-rich groundwater.*

**Response:** Thank you. We added a paragraph on how a confluence changes main-stem conductance: "Where a tributary joins the main stem, the conductance downstream of the junction is approximately the discharge-weighted average of the two inflowing waters, a two-end-member mixing relation of the kind used to separate flow components from tracer concentrations" {L106} (Genereux, 1998). It sets up the comparison between tributary and fault boundaries in the Results. We agree that dilution may be partly offset by salt-rich groundwater, but our data cannot test it: "Dilution by a tributary could also partly mask a groundwater input at the same location, which we cannot separate without discharge measurements." {L112}

**Comment R2.2**

*Methods. Clarify wheter the normalization was conducted across the data from the whole site or within each zone in each site.*

**Response:** On review, the normalization was not used in any of the statistical tests, so the question no longer applies. We removed it from the Methods and the figures now show unnormalized values. See our response to R1.9.

**Comment R2.3**

*On pg 4 you say – 'within-zone scatter is small relative to the between zone contrasts'. Show the standard deviation values to demonstrate this here and throughout the methods and figures (i.e., mean values should have +/- standard error or std, whichever is more statistically appropriate). In addition, somewhere you should show the actual data – if not in a new table 2 in the main text, then at least in an appendix.*

**Response:** Agreed. Zone means are now mean ± standard deviation in the text, Table 1 and the zone labels of Figures 2–4. We removed the quoted statement because, once zones were restricted to single days (R1.7), the comparison no longer held in that form. The text now gives the values: "Within a single day the zone standard deviations are 0.1–2.6 $\mu$S/cm." {L294} All individual measurements (with date, distance along channel, position, temperature, zone assignment and whether the point is a fault point, located downstream of a fault) are in the new Appendix A, which "lists every specific-conductance measurement from the 2018 survey" {L873}, and in the public data repository.

**Comment R2.4**

*Results. Pg 5 – It looks like in Fig 3. the fault crosses the stream at several locations, not just where you've mapped in the contrast between two zones. Additional justification/explanation for how the zones (and contrast) were chosen given these additional fault crossings would be helpful.*

**Response:** You are right that the fault line in Figure 3 meets the stream in several places. It comes from the county fault map in Budnik et al. (2010; Fig. 3.10, printed at about 1:340,000), whose authors describe the fault locations as "largely conjectural since the faults themselves are rarely exposed at the surface" {L147} (p. 18). The apparent multiple crossings are therefore not reliable, and we removed the crossing markers from Figure 3.

The boundary between Zones 2-1 and 2-2 (Contrast B, about 4.0 km along the survey) is our field interpretation of the fault position: an abrupt, roughly 90° bend in the channel, which we took to be the creek turning to follow the fault. Section 2.2 now says so: "At Site 2 the digitized county-scale trace runs along the creek for the whole surveyed reach, so it cannot resolve where the fault crosses the channel." {L199} It also states the limit: "An abrupt bend is consistent with structural control of the channel, but it does not by itself show that groundwater flows along the fault there." {L202} Section 3.1 adds that the trace intersections at about 1.0 and 5.8 km along the survey "are not reliable fault positions, so we did not use them" {L321}. Section 2.2 also explains the fault points ringed at the upstream end of Zone 2-1 in Figure 3: they lie where the mapped fault zone first meets the creek, and "These upstream points do not define a second contrast: Site 2 has a single contrast, B, at the bend." {L210} Because a bend does not show groundwater flow, and the change-point analysis places the onset of the Site 2 increase about 1.6 km upstream of the bend (R1.5(3)), Contrast B is reported as an observation worth testing, not as a step at a fault.

**Comment R2.5**

> *The increase in conductance across contrast B is very small – what is the uncertainty in the YSI meter? What was the variability in the readings in each zone? I have seen conductance values change more than 5 us/cm by moving across a stream or from the surface to a deeper point in the water. Without seeing the raw data variability I am skeptical this small difference represents an abrupt change in water source, even if found to be statistically significant after normalization.*

**Response:** We share your caution. Specific conductance was measured with a Thermo Scientific Orion 013010MD cell, not a YSI instrument (YSI was the 2020 chloride sensor). The accuracy, and why a common calibration offset cancels in a within-day difference, are in our response to R1.11 (about $\pm 1.7\,\mu\mathrm{S\,cm^{-1}}$ at Site 2). Within-zone standard deviations at Site 2 are 1.1 and $1.9\,\mu\mathrm{S\,cm^{-1}}$ (Table 1), all readings are in the Appendix, and the tests use unnormalized values.

You are right about cross-channel and depth variability. All readings were taken in the thalweg, so we cannot rule out that it contributes to a difference of this size: "Each reading is a single thalweg value, with no repeats across the channel or at depth, so lateral and vertical variation, which could exceed the small contrasts, is unmeasured." {L635} We also agree the change is not abrupt (R1.5(3)), and describe Contrast B as a small, gradual change near the resolution of the survey that does not demonstrate a point input at the fault. The Site 2 survey may also not represent baseflow: "Flow was very high for the duration of the Site 2 survey, and we canoed part of it, so this survey may not represent baseflow." {L323}

**Comment R2.6**

> *I am most convinced by contrast E – although by looking at figure 4, the change in conductance value actually occurs within zone 3-3. It's possible that the fault location is slightly off, correct? If so, could you show that with a fault zone rather than a single line, or at least discuss the possibility that the fault trace is actually east of where it's mapped?*

**Response:** Thank you for this careful reading, which together with R1.7 led us to find a presentation mistake. The jump you see inside Zone 3-3 falls exactly where the 19 October survey ends and the 26 October survey begins: it is the day-to-day offset described in R1.7, not a change along the channel. Within a single day, the change across the mapped fault is $+1.6\,\mu\mathrm{S\,cm^{-1}}$ (95% CI $-0.3$ to $+3.4$, $p = 0.090$) and not significant, so we no longer present Contrast E as the strongest

evidence.

On the fault position: within the 26 October survey alone, conductance rises by about 4 $\mu$S cm$^{-1}$ over the first ∼190 m of the reach and then stays near 360 $\mu$S cm$^{-1}$, including where the channel crosses the digitized trace about 520 m along it. The change-point breakpoint is at 194 m: "This breakpoint sits at the three-point minimum segment the method allows, 330 m upstream of where the channel crosses the digitized trace (at 523 m), 123 m in straight-line distance from the nearest trace, and 522 m upstream of the Contrast E boundary." {L401} We agree the traces are uncertain and looked for a published accuracy with which to draw a zone, but found none we could cite (see R2.4), so we draw lines rather than an error zone we cannot support. On the printed scale of the source figure, 123 m is less than 0.4 mm, so the trace could indeed lie slightly off its mapped position, as you suggest, but our data cannot test this.

The rise is also not robust: it is not significant as a zone contrast, and it could reflect an input unrelated to the fault. The manuscript does not attribute it to the fault: "Contrast E is within the instrument's stated accuracy and its only breakpoint lies near the start of a short transect, 330 m upstream of where the channel crosses the trace." {L469} The Limitations add that we "cannot say how far a real fault lies from any mapped position" {L648}.

**Comment R2.7**

> *The chloride measurements could be tested for statistically significant difference above and below contrast E – that would be interesting to see as well.*

**Response:** Thank you for this suggestion. Locating the Contrast E fault points on the chloride transect from their GPS positions, we found that the transect ends just downstream of them, so only 2 of the 18 main-stem samples lie downstream, too few for a two-sample test. The Results now say: "Near Contrast E, chloride is low and flat: only 2 of the 18 samples lie downstream of the Contrast E fault points (13.75 and 15.0 mg/L), and the last five samples are 13.75–17.5 mg/L. The transect therefore shows no chloride increase at Contrast E. With so few points downstream of E we did not attempt a two-sample test." {L418}

This corrects the original manuscript, which attributed the chloride peak to the Contrast E fault. The transect "ran 2,351 m from Z to Z′" {L407}, mostly within Zone 3-3. The three highest values (50, 36.25 and 40 mg L$^{-1}$) lie "133–468 m from the nearest Contrast D fault point and 293–391 m from the nearest digitized fault trace" {L409}. To describe where chloride changes without two-sample tests, we applied the same change-point analysis used for conductance: "The segmented regression selects one breakpoint at 467 m (bootstrap interval 308–800 m)" {L413}, about 125 m from the nearest Contrast D fault point. Because it is driven largely by one 50 mg L$^{-1}$ sample, we report it as an observation worth testing: "We do not treat the chloride peak as support for the conductance result at any contrast." {L476} The chloride survey (July 2020) was made in a different year from the conductivity survey, so we do not combine the two.

**Comment R2.8**

> *In Zone 3-1 it looks like there are two distinct sets of data overlapping on the figure – a set of higher ( 370) conductance measurements and a sect of lower ( 340) conductance measurements. This should be further explained or clarified in the figure caption and main text.*

**Response:** Correct: these are measurements of the same reach on two days (19 and 26 October

2018), differing by the day-to-day offset described in R1.7. Figure 4 now distinguishes the days by symbol, with an inset for the overlapping reach, and the caption and text explain it: "Zones 3-1 and 3-2 were surveyed on both days (Figure 4, inset): Zone 3-1 averaged 340.1 ± 0.4 $\mu$S/cm ($n = 6$) on 19 October and 370.5 ± 2.6 ($n = 11$) on 26 October" {L373}.

**Comment R2.9**

*Discussion. While some alternative explanations for the increase in conductance at contrasts B and E are discussed in 4.2, landuse differences are dismissed because all sampling was within a conserved forest – however, the maps show distance to roadways and density of roads varies around the various sites and zones within sites. Whether or not the differences in roadway density (and road salting practices) could affect the specific conductance values should be discussed – especially when the observed differences at contrast B are so small. Although you sample in summer, you mention in your introduction that salt can persist in soil for up to 5 months.*

**Response:** Agreed; see our response to R1.12, including the withdrawn conserved-forest argument and the road-distance medians (Table 2). Section 4.2 now notes that the distances do not capture drainage or road density, and that they "say nothing about salt stored in roadside soils and shallow groundwater, which may be flushed to the channel late in the season and which our survey cannot distinguish from deeper, fault-routed water" {L523}. We removed the soil-persistence sentence from the Introduction (see R1.2), but the underlying point stands for roadside soils. Our survey was in late September and October 2018, not summer, and the text now says so {L166}: "Our surveys were made in late September and October, so a late-season release of such stored salt is a possibility we cannot exclude" {L525}.

**Comment R2.10**

*Figures and Tables. Figure 1 (b) is difficult to see details - I would suggest simplifying so just major streams and faults are shown; remove other extraneous details. Or at least highlight the streams in question.*

**Response:** Done. Figure 1 is now a single panel showing only major streams, highways and the mapped faults on a light basemap, with Wappinger Creek highlighted, the three study sites boxed, and wastewater facilities marked. The county fault map that formed panel (a) is no longer reproduced; the faults are digitized from it and cited.

**Comment R2.11**

*Figure 2: Label contrast A. Add flow direction.*

**Response:** Done: Contrast A is labeled and a flow arrow is added.

**Comment R2.12**

*Figure 3: Label contrast B. Add flow direction.*

**Response:** Done: Contrast B is labeled and a flow arrow is added. The figure also marks the nearby wastewater facilities, and the Contrast B box labels the boundary as the fault position interpreted in the field (see R2.4).

**Comment R2.13**

*Figure 4: Label each contrast E, D, C. Add flow direction. Add margin of error for fault location if known. Add line Z-Z'. Offset or include an inset to better show the overlapping points in Zone 3-2 and 3-1.*

**Response:** Done. Contrasts C, D and E are labeled and flow direction is shown. We found no published margin of error for the fault locations that we could cite (see R2.4 and R2.6), so the faults are lines without an error zone, and their uncertainty is discussed in the text and Limitations. The chloride transect Z–Z′ is drawn from its GPS positions (yellow band). Survey days are distinguished by symbol (circles and triangles), and an inset enlarges the overlapping reach in Zones 3-1 and 3-2.

**Comment R2.14**

*Figure 5: Either add line Z-Z' in figure 4 or add an inset figure to Fig 5 to show the location of this transect relative to site 3 in the study*

**Response:** Z–Z′ is now shown in Figure 4. Figure 6 marks the positions of the Contrast D and E fault points, the trace crossing and the Kelly et al. (2019) monitoring site along the transect, together with the change-point fit.

**Comment R2.15**

*Table 1. Show the normalized mean values for each zone. Show some measure of variance for each zone. Consider including chloride measurements above and below contrast E as well*

**Response:** Table 1 now reports each zone's mean ± standard deviation (see R1.6). We respectfully decline to add normalized means: the normalization has been removed (R1.9) and all tests use specific conductance in $\mu\mathrm{S\,cm^{-1}}$, so the table reports unnormalized means. Too few chloride measurements lie downstream of Contrast E for a two-sample comparison (R2.7), so the chloride results are reported through the change-point analysis and Figure 6 instead.

## Response to Reviewer 3

**Comment R3.1**

*Minor Revisions*

**Response:** Thank you for the assessment. The revised manuscript addresses the comments of Reviewers 1 and 2, and we hope it also meets your expectations.